

Creation of Link files

MODULE 2: FILE OPERATIONS, LINKS, AND DATA PROCESSING

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Introduction

In Linux, we can create links between files using the **ln** command.

Every file has two parts: the actual data blocks stored on the disk, and an inode (index node), which is a pointer holding the file's metadata (size, permissions) and pointing to those data blocks.

There are two ways to create a link to a file in Linux:

- **Symbolic Links (Soft Links)**, which act like shortcuts, and
- **Hard Links**, which act as direct clones sharing the same underlying data (inode).

Symbolic Links (Soft Links)

A symbolic link is a special file that points to the path of another file or directory. If the original file is deleted, the symbolic breaks.
To create a symbolic link, use the **–s** flag: **ln –s original_file.txt link_file.txt**

Example for a file: ln -s original.txt soft_link.txt
Example for a directory: ln -s /var/www/html/ my_web_folder

```
yoges@LAPTOP-DCISH7AJ:~/linux-fundamentals/creation_of_link_files$ ls -l
total 12
-rw-r--r-- 2 yoges yoges   26 Nov  1 15:10 hard_link.txt
-rw-r--r-- 2 yoges yoges   26 Nov  1 15:10 original.txt
lrwxrwxrwx 1 yoges yoges    9 Nov  1 15:21 soft_dir_link -> test_dir/
lrwxrwxrwx 1 yoges yoges   12 Nov  1 15:13 soft_link.txt -> original.txt
drwxr-xr-x 2 yoges yoges 4096 Nov  6 04:50 test_dir
```

The Link Behaviour: The l at the very beginning of the permission string confirms it is a link. The arrow explicitly reveals soft_link.txt -> original.txt.

The Broken Link: If you rename or delete original.txt, the soft link will turn red/blink in your terminal. It breaks because the file name it points to no longer exists, creating a "dangling link."

Hard Links (The Dual Identity)

A Hard Link creates a new filename that points directly to the exact same i-node as the original file. It behaves exactly like a copy, but it occupies no additional disk space because both names point to the same physical data block.
To create a hard link, omit the **–s** flag: **ln original_file.txt hard_link.txt**

How to verify: Check Inodes, Run **ls -i** to print the inode numbers. **ls -i original.txt hard_link.txt**

Limitations: Hard links cannot link directories on most systems and cannot span across different filesystems/partitions.

